

Effects of Electrodes on Space Charge in Polypropylene

E. Nakane^{1*}, K. Kaneko¹, T. Mori¹, T. Mizutani¹, H. Takino² and M. Ishioka²

¹Department of Electrical Engineering and Computer Science, Nagoya University, Japan

²Japan Polyethylene Co. Ltd., Japan

Abstract: We investigated the effects of electrodes, i.e. Al plate, semiconducting (SC) layer and evaporated Al, on space charge in polypropylene (PP) by using the pulsed electroacoustic (PEA) method and also on conduction current. Evaporated Al electrodes suppressed the carrier injection and, as a result, space charge compared with Al plate or SC layer (mechanically contacted electrode). The conduction current also revealed that the carrier injection was suppressed by evaporated Al. The space charge formation and the current showed a significant dependence on not only electrode material but also contact condition between metallic electrode and insulator.

Introduction

The space charge accumulation sometimes affects electrical conduction and breakdown phenomena because it enhances a local electric field. The space charge formation in an insulating polymer depends on a lot of factors such as temperature, electric field, polarity of applied voltage, electrode and additive [1]. Many papers have been published on their effects on space charge behaviors in polyethylene (PE) which has been widely used as an insulating material for power cables.

We have investigated space charge dynamics in polypropylene (PP) which has recently attracted much attention as cable insulation because of its high melting point and suitability of recycle. For the Al plate and semiconducting (SC) layer electrode system, we reported that charge carriers were injected from both electrodes to form homo space charges in PP at 40°C [2]. In this paper, we investigated the effect of electrodes, Al plate, SC layer and evaporated Al, on space charge and conduction current.

Experimental

We used 100 μm -thick PP film (density 0.886 g/cm^3 , melting point 125°C) which was polymerized with metallocene catalyst [3]. The sample is a random copolymer with 3 wt% of ethylene. By using different combinations of Al plate, SC layer and evaporated Al, we prepared the four kinds of electrode systems, (1) Al plate and SC layer (no evaporated Al), (2) evaporated Al on both sides, (3) evaporated Al on the side of

Table 1: Electrode system

Name	Electrode system
No evaporated Al	Al / PP / SC
Evaporated Al (SC layer side)	Al / PP / <u>Al</u> / SC
Evaporated Al (Al plate side)	Al / <u>Al</u> / PP / SC
Evaporated Al (both sides)	Al / <u>Al</u> / PP / <u>Al</u> / SC

Al : Al plate Al : evaporated Al

grounded Al plate and (4) evaporated Al on the SC layer (see Table 1). Al denotes evaporated Al in Table 1. Al electrode was evaporated under a vacuum $< 3 \times 10^{-5}$ Torr and its thickness was about 20 nm or 100 nm.

Space charge distributions were measured with the PEA method [4] as shown in Fig. 1. A sample was sandwiched between a grounded Al plate and a SC layer. A positive or negative DC voltage was applied to SC electrode. They are called 'positive polarity' and 'negative polarity', respectively. The DC field of 50 MV/m was applied for 90 minutes and then the electrodes were short-circuiting for 30 minutes. The space charge profiles were measured every minute at 40°C, where the amount of space charge was the largest. The measurement temperature was controlled by putting the PEA electrode system in an oven. Charging currents were also measured at the same conditions.

Results and discussion

Evaporated Al on both sides

Figure 2 shows the space charge distributions in PP with electrode systems of (a) -Al/PP/SC+ (mechanically-contacted Al plate and SC layer, no evaporated Al) and

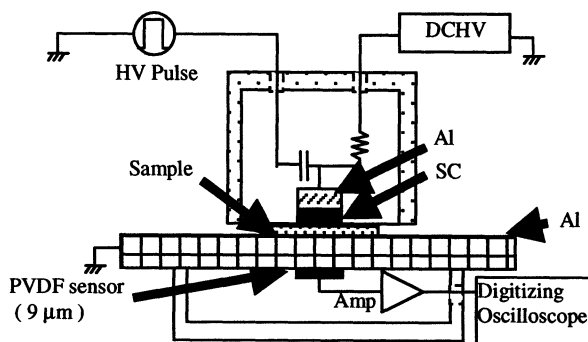


Figure 1: Setup for space charge measurement.

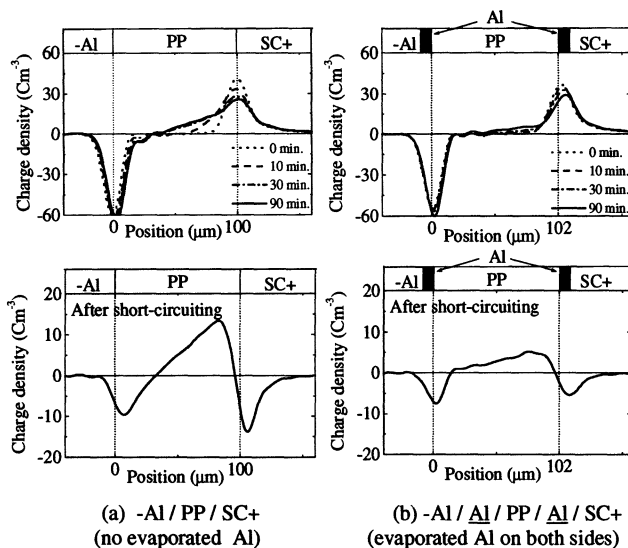


Figure 2: Space charge distributions in (a) -Al / PP / SC+ and (b) -Al / Al / PP / Al / SC+ under applied field and after short-circuiting, where Al is 20 nm-thick evaporated Al.

(b) -Al/Al/PP/Al/SC+ (evaporated Al on both sides). The applied voltage was positive polarity. For no evaporated Al (Fig. 2(a)), positive and negative homo space charges are formed near the SC anode and Al cathode, respectively. The former is larger than latter. For evaporated Al on both sides (Fig. 2(b)), on the other hand, the amounts of space charge obviously decrease and only positive charge is observed. These suggest that both positive and negative carrier injections are suppressed by evaporating Al.

Figure 3 shows space charge distributions for negative polarity. In case of no evaporated Al, you can see a large positive homo space charge near the Al anode (Fig. 3(a)). Space charge is very small for evaporated Al on both sides and a small amount of negative homo space charge is observed near the SC cathode as shown in Fig. 3(b).

From Figs. 2 and 3, we obtained the following conclusions; (1) Positive homo space charge is dominant in Al/PP/SC electrode system, (2) Evaporated Al suppresses the carrier injection compared with mechanically-contacted Al plate or SC layer, and (3) a small amount of space charge is observed near the SC electrode for evaporated Al on both sides, but not near the Al plate.

In order to clarify the reason why a small amount of space charge is observed near the SC layer in case of evaporated Al on both sides, we prepared 100 nm-thick evaporated Al electrodes on both sides of a PP film. The resultant space charge distribution is shown in Fig. 4. In this case, there are almost no space charges in PP. The similar result was obtained for PP with evaporated Al

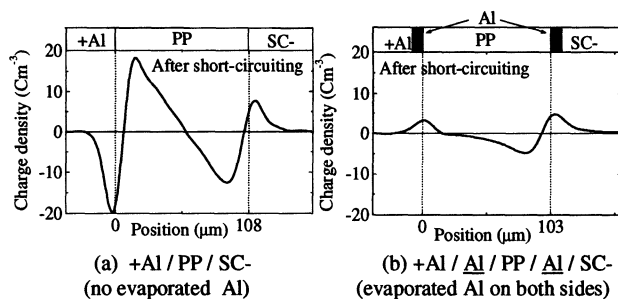


Figure 3: Space charge distributions in (a) +Al / PP / SC- and (b) +Al / Al / PP / Al / SC- after short-circuiting, where Al is 20 nm-thick evaporated Al.

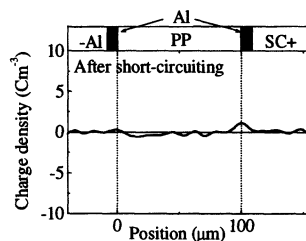


Figure 4: Space charge distributions in -Al / Al / PP / Al / SC+ after short-circuiting, where Al is 100 nm-thick evaporated Al.

on both sides in negative polarity. It suggests that 20 nm-thick evaporated Al is a little too thin to prevent the charge injection from the neighboring SC electrode. Some damages is considered to be introduced in 20 nm-thick evaporated Al adjacent to a flexible SC layer by the mechanical stress during the PEA measurement. Through them, some carriers may be injected from the SC layer. This is a possible reason for the effect of thickness of evaporated Al.

Evaporated Al electrode on one side

Figures 5 to 8 show the space charge distributions in PP with evaporated Al on one side, under short-circuiting after the applied voltage for 90 minutes. The thickness of evaporated Al was about 20 nm. Figure 5 shows the result of PP with evaporated Al on the SC side for positive polarity. The positive space charge is not observed. The magnitude of negative charge injected from the Al cathode is larger than that of no evaporated Al (Fig. 2(a)), and it is extended toward the SC anode because of the suppression of hole injection from the anode. For negative polarity, on the other hand, only positive space charge is accumulated throughout a sample as seen in Fig. 6. Figures 7 and 8 show the space charge profiles in PP with evaporated Al on the Al plate side. For positive polarity (Fig. 7), positive charge injected from the SC electrode is observed across a sample. Only negative charge injected from the SC is found in PP for negative polarity as shown in Fig. 8,

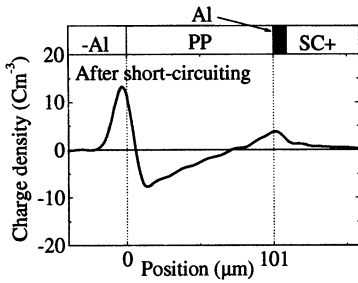


Figure 5: Space charge distributions in -Al / PP / Al / SC+ after short-circuiting, where Al is 20 nm-thick evaporated Al.

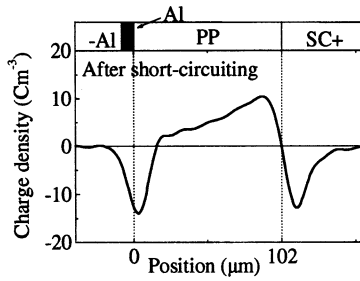


Figure 7: Space charge distributions in -Al / Al / PP / SC+ after short-circuiting, where Al is 20 nm-thick evaporated Al.

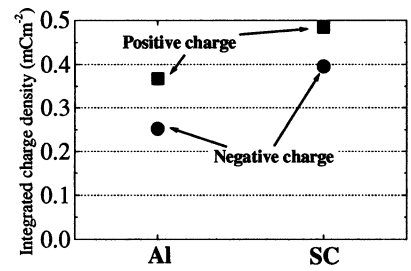


Figure 9 : Amounts of space charge after application for 90 minutes.

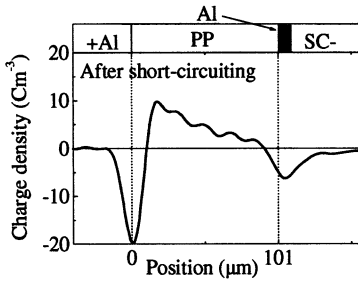


Figure 6: Space charge distributions in +Al / PP / Al / SC- after short-circuiting, where Al is 20 nm-thick evaporated Al.

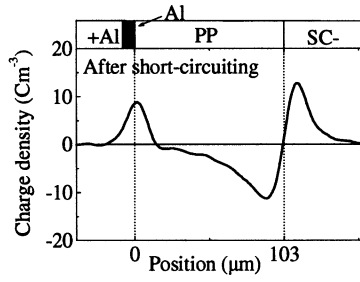


Figure 8: Space charge distributions in +Al / Al / PP / SC- after short-circuiting, where Al is 20 nm-thick evaporated Al.

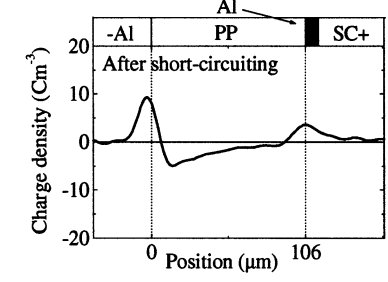


Figure 10: Space charge distributions in -Al / PP / Al / SC+ after short-circuiting, where Al is 100 nm-thick evaporated Al.

vice versa. In PP with evaporated Al on one side electrode, only charge carriers injected from the non-evaporated Al plate or SC layer are observed throughout the bulk, but not from evaporated Al.

Figure 9 shows integrated space charge densities on the Al and SC sides obtained from Figs. 5 to 8. From the result, it is found that carriers (electrons or holes) are injected more easily from the SC layer than from the Al plate, and that holes are injected more easily from both SC layer and Al plate than electrons. The authors [1] and Chen et al. [5] reported similar electrode effects on space charge in LDPE.

In PP with 100 nm-thick evaporated Al on the SC side, negative homo space charge is formed across a sample as shown in Fig. 10. It is similar to that in Fig. 5, except the vicinity of the SC layer. In Fig. 10, negative space charge reaches the anode because of the suppression of hole injection from the anode (100 nm-thick evaporated Al). On the other hand, a small hole injection from 20 nm-thick Al/SC anode is observed in Fig. 5. But its amount is negligible compared with the injection from non-evaporated SC anode (Fig. 2(a)).

Charging currents

Figure 11 shows charging currents in PP with the four electrode systems for (a) positive polarity and (b) negative polarity. The thickness of evaporated Al was

20 nm in this experiment. As you can see in Fig. 11, the current in PP is suppressed by evaporated Al electrode for both polarities. It is reduced to a third by evaporated Al on one side. The current in PP with evaporated Al on both sides is by about one order smaller than that in PP with no evaporated Al. This is also a direct evidence for the suppression of carrier injection by evaporated Al.

For positive polarity, the current decays more rapidly in PP with evaporated Al on the SC side. Figure 12 shows the space charge distributions in PP with evaporated Al on the SC side under the field of positive polarity. Negative carriers are injected from the Al cathode and reach the counter electrode within 10 minutes. The space charge profile becomes almost steady-state within about 30 minutes. Such space charge dynamics may affect the current transient and result in the rapid current decay in PP with evaporated Al on the SC side in Fig. 11(a). Furthermore the current in PP with evaporated Al on both sides for negative polarity is higher than that for positive polarity. It may suggest that electrons are more mobile than holes, considering that the current is controlled by carriers passing through the 20 nm-thick evaporated Al.

The results of charging currents are closely related to those of space charge. Their combination will give us further information about charge transport and space charge behavior in PP.

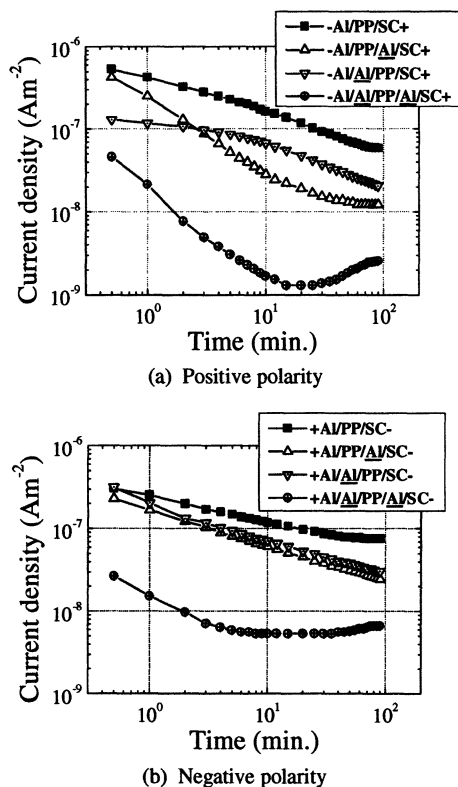


Figure 11: Total current density vs. time for various electrode systems under applied field (a) in positive polarity and (b) in negative polarity. Evaporated Al had the thickness of 20 nm.

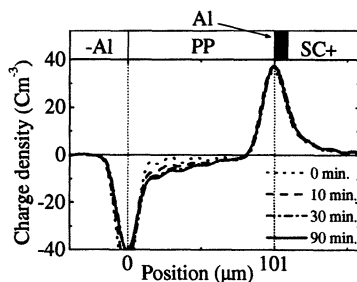


Figure 12: Space charge distributions in -Al plate / PP / Al / SC+ under the applied field. Evaporated Al had the thickness of 20 nm.

Conclusions

The effects of electrodes on space charge characteristics in PP (random copolymer with ethylene) have been investigated. Both positive and negative carriers were injected from mechanically-contacted Al and SC electrodes. However, evaporated Al electrode remarkably suppressed the carrier injection and, as a result, space charge in PP. Although a small leakage injection from the SC electrode passing through 20 nm-

thick evaporated Al was observed, it was prevented by using 100nm-thick evaporated Al.

The results of space charge in PP revealed that both holes and electrons are injected more easily from SC electrode than Al one. Furthermore, holes are injected more easily in PP in the field region up to 50 MV/m than electrons. From a temporal change in the space charge formation, it is concluded that electrons are more mobile in PP than holes.

The space charge was affected not only by electrode materials but also by contact conditions. Both electron and hole injection from evaporated Al electrode were much suppressed compared with those from mechanically-contacted Al. Its mechanism is now under consideration.

The results of space charge were closely related to those of currents in PP. Their combination gave us further information about carrier transport and space charge in PP.

References

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Author address: Emi Nakane, Department of Electrical Engineering and Computer Science, Nagoya University, Furo-cho, Chikusa, Nagoya 464-8603, Japan, Email: m_nakane@echo.nuee.nagoya-u.ac.jp.